

DATA STRUCTURES - Examination 2018

Total Marks: 70

SECTION A

1.

Attempt all questions in brief.

- (a) Define data structure and its types.
- (b) What is a stack overflow condition?
- (c) State the time complexity of bubble sort.
- (d) Define a priority queue.
- (e) What is a complete binary tree?
- (f) Define a spanning tree.
- (g) What is linear probing?

SECTION B

2.

Attempt any three of the following:

- (a) Explain stack and its applications with infix to postfix conversion.
- (b) Explain linked list types with insertion and deletion algorithms.
- (c) Explain heap sort with an example.
- (d) Explain graph representation using adjacency matrix and adjacency list.
- (e) Explain hashing and collision resolution techniques.

SECTION C

3.

Attempt any one part of the following:

- (a) Explain binary search tree with insertion and deletion examples.
- (b) Explain Prim's and Kruskal's algorithm for minimum spanning tree.

4.

Attempt any one part of the following:

- (a) Write the binary search algorithm and explain its complexity.
- (b) Explain AVL tree rotations with examples.